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Inventor(s)

Rabe; Philip et al.

Electric Heating Device

Abstract

An electrical heating device for a motor vehicle includes a housing which encloses a heating chamber which is coupled in a thermally conductive manner to an electrical resistance element and which encloses a control device. The electric heating device further has a pressure equalizing device for equalizing a pressure difference between the interior of the housing and the environment of the housing. The pressure equalizing device is sealingly inserted in a receiving bore which is formed to receive the pressure equalizing device and which is recessed in the housing.

Inventors: Rabe; Philip (Neustadt an der Weinstrasse, DE), Terlinden; Felix (Reilingen, DE)

Applicant: Eberspächer catem GmbH & Co. KG (Herxheim, DE)

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to an electric heating device.

2. Discussion of the Related Art

[0002] An electric heating device known from EP 3 334 242 A1, the subject matter of which is incorporated by reference by way of background information, is considered to be of the same general type.

[0003] In the aforementioned prior art, the one electrical heating device has a control housing that is connected to the plug-in contact with contact elements for the power current or to the plug-in contact with contact elements for control signals with corresponding interfaces. These interfaces also allow gas exchange between the inside of the housing and the environment and thus pressure equalization.

[0004] On this basis, the present invention seeks to provide an electrical heating device which can be adapted to different requirements in an improved manner.

SUMMARY

[0005] To solve this problem, the present invention proposes an electrical heating device, typically configured for use in a motor vehicle, which comprises a housing and a pressure equalization device. The housing encloses a control device and also encloses a heating chamber which is coupled in a thermally conductive manner to an electrical resistance element, the housing enclosing a control device. The pressure equalization device is configured to equalize a pressure difference between the interior of the housing and the environment of the housing. The pressure equalization device is inserted in a sealing manner in a receiving bore which is designed to be adapted to receive the pressure equalization device and which is recessed in the housing.

[0006] In the electrical heating device according to the invention, the housing, which is usually formed of metal at least with its housing part surrounding the control device, may have a receiving bore which passes through a wall of the housing and accordingly connects the interior of the housing, in particular the part of the housing receiving the control device, to the environment. This locating hole may be specially adapted to accommodate a pressure equalization device.

Accordingly, the mounting hole has a dimension that is precisely adapted to accommodate the pressure equalization device. The pressure equalization device is generally used solely to equalize a pressure difference between the inside of the housing and the surroundings of the housing.

[0007] In this way, the electrical heating device according to the invention enables pressure equalization between the interior and exterior, for example, due to temperature changes.

[0008] According to a possible further development of the present invention, the pressure equalization device may be screwed into the receiving bore, for which purpose the pressure equalization device, in particular a pressure equalization plug thereof, is generally provided with an external thread and the receiving bore with an internal thread. Alternatively, the pressure equalization device can also be mounted in the locating hole in the form of a clip. For this purpose, the pressure equalization plug can be inserted into the locating bore and provided with locking tabs on its underside, for example, which protrudes into the housing.

[0009] The pressure equalization plug may be penetrated by a pressure equalization channel, which usually extends centrally within the receiving bore in the longitudinal direction thereof and is closed by an air-permeable but water-impermeable device. This device is used to retain moisture, in particular water in liquid or vapor form. This device can be made of metal or plastic, in particular as a membrane that provides a different permeability for gases and moisture. Such properties can also be provided by filters made of plastic, natural material such as cellulose fibers, and/or metal.

[0010] The device may be provided on the outside of the pressure compensation plug, and may be connected to the front of this pressure compensation plug. For this purpose, the pressure compensation plug may have a radial flange that is wider than the rest of its outer diameter, on the

front of which the device can be applied, in particular in the form of a diaphragm, and connected to it with a material bond. On the side opposite the device, there is usually a sealing ring between the radial flange and the housing, via which the pressure equalization device is sealed off from the housing.

[0011] Instead of a separate sealing ring, it is also possible to form a seal as an integral part of the pressure compensation plug, for example by overmolding the pressure compensation plug with a soft elastic plastic.

[0012] The membrane can be flexible in order to compensate for certain pressure fluctuations within the control housing through elastic deformation. More typically, however, the diaphragm is permeable to air and impermeable to water, so that air can escape from the control housing in the event of any overpressure or air can flow into the control housing through the diaphragm in the event of a certain negative pressure, so that pressure equalization is achieved within the control housing. Simple systems with only one membrane or multi-part systems with a multilayer film structure can be used for venting, whereby different layers of the film structure can have different functions such as a mechanical function or a retaining function. A sintered filter can also be used as a mechanism via which of ventilation through the pressure equalization channel may be achieved.

[0013] The membrane can be welded, such as ultrasonically welded, to the pressure equalization plug. Alternatively or additionally, the diaphragm can be bonded to the pressure equalization plug.

[0014] It has been found that an air-permeable but liquid-impermeable membrane may be made of ePTFE or PTFE. This membrane should have a thickness between 120 μm and 240 μm , in particular 180 μm $\pm$ 50 μm . Practical tests by the applicant have shown that such a membrane prevents water or water vapor from entering the control housing, but air can escape from the control housing through the membrane.

[0015] A cover cap may be provided on the outside of the housing and usually covering the diaphragm, which is connected to the pressure equalization plugs, such as being latched or pressed.

[0016] This cover cap can also be formed integrally on the pressure compensation plug, for example on a uniform injection-molded component, while retaining a compensation channel within the pressure compensation plug. In such a variant in particular, the diaphragm may be located at the end of the pressure compensation plug that protrudes into the housing and can be connected there to the pressure compensation plug in the manner described above.

[0017] This cover cap and/or the pressure equalization plug can be made of plastic or metal. It is understood that a compensation gap for pressure equalization purposes may be provided between the cover cap and a frontal opening to the pressure equalization channel, which is usually covered by the diaphragm.

[0018] The housing may be thickened in the area of the locating bore, whereby the locating bore is provided with sufficient length for a firm connection with the pressure equalization device, in particular the pressure equalization plug inserted into the locating bore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Further details and advantages of the present invention can be seen from the following description of an embodiment in conjunction with the drawing. In this show:

[0020] FIG. 1: An exploded perspective view of an embodiment of the present invention;

[0021] FIG. 2: an enlarged perspective top view of a part of the housing according to Figure comprising the pressure equalization device;

[0022] FIG. 3: a longitudinal sectional view along line III-III as shown in FIG. 2.

DETAILED DESCRIPTION

[0023] FIG. 1 shows an electrical heating device 2 with a housing base part 4, which is connected

to a heating chamber cover, not shown here, on its lower side in FIG. 1 to form a first heating chamber **6.1**. Reference sign **10** indicates a fluid housing which is accommodated in a receiving space **12** formed by the housing base part **4** and forms a second heating chamber **6.2**. In this receiving space **12**, which is bounded on the underside by a base **14** and on the edge by a circumferential wall **16** extending from the base **14**, and between the base **14** and the fluid housing **10**, there is a first insulating layer **18**, which can be placed against the base **14**, a first contacting layer **20** and a heating device **22** with a plurality of electrical resistance elements **23**, in the present case in the form of PTC elements **24**, and a positioning frame **26** with receptacles **28** for receiving the PTC elements **24**.

[0024] On the side of the heating device **22** opposite the first contacting layer **20**, a second contacting layer **30** is provided, on the side of which opposite the heating device **22** a second insulating layer **32** is arranged. The PTC elements are therefore electrically conductive against the contacting layers **20**, **30** and are energized via these. The PTC elements **24** are ceramic cuboid components, which are provided with a metallization on opposite main side surfaces for current conduction. The main side surface is the surface of the cuboid with the largest surface area. The main side surfaces are connected to each other by edge surfaces, which determine the height of the PTC elements and do not have any metallization. The main side surfaces are generally each larger by a factor of 5 than one of the edge surfaces, typically than the sum of all the edge surfaces.

[0025] The second insulating layer **32** is formed as a prestressing device **33** by a silicone film, which is able to absorb certain compressions by elastic deformation and thus apply the electrical heating device **22** between the housing base part **10** and the housing base **14** under prestress against the housing base part **10** and the housing base **14**.

[0026] The layering of the first insulating layer **18**, the first contacting layer **20**, the heating device **21**, the second contacting layer **30** and the second insulating layer **32** is hereinafter also referred to as a layered structure **34**.

[0027] On the side of the fluid housing **10** facing away from this layered structure **34**, a transistor insulation **40** is located between a printed circuit board **38** forming a control device **36** and the fluid housing **10** in the embodiment example shown. Reference sign **42** identifies a housing cover which is connected to the housing base part **4** to form a housing identified by reference sign **44**. The housing base part **4** and the housing cover **42**, and possibly other housing parts, are of shielding design, i.e. made of metal and/or provided with a separate shielding on the inside, outside or in the walls of the respective housing part, which may be made of plastic. A possible metallic material is aluminum or stainless steel for corrosion resistance.

[0028] A power connector **46** and a control connector **48** are shown in FIG. 1 on the underside of the housing base part **4** opposite this housing cover **42**. These two plugs **46**, **48** are connected to the housing base part **4** in a sealed manner and have various male electrical plug contacts which are led through the respective housings of the plugs **46**, **48** in a sealed manner and are plug-contacted in the printed circuit board **38** and are electrically connected to conductor tracks of the printed circuit board **38** via this plug contacting. Details of this are described below in connection with FIG. 4. For the plug-in contact, the printed circuit board **38** has female contact tongue receptacles, which are described in EP 2 236 330 A1.

[0029] Furthermore, inlet and outlet nozzles **50**, **52** are provided on the underside for the connection of pipes or hoses carrying the fluid to be heated. Reference sign **53** indicates the sealing arrangement **53** shown in FIG. 1 below the fluid housing **10**, which in the present case is formed by two sealing rings **54**. Reference sign **56** indicates screws for fixing the fluid housing **10** relative to the housing base part **4** with the layered structure **34** interposed. This screw connection places the layers of the layered structure **34** under preload against the fluid housing **10** and the base **14** of the housing base part **4**.

[0030] FIGS. 2 and 3 show details of a pressure equalization device **58**, which is arranged adjacent to the power connector **46** on the housing base part **4**. For this purpose, the housing base part **4** has

an elevation **60**, by which the wall thickness of the housing base part **4** is increased (see FIG. 3). This elevation **60** is provided with a receiving bore **62**, which has an internal thread to which an external thread of a pressure equalization plug **64** is screwed. The pressure equalization plug **64** is penetrated by a pressure equalization channel **66** extending in the longitudinal direction of the receiving bore **62** and has a radial flange **68** provided on the outside of the elevation **60**, which receives a sealing ring **70** between itself and an end face of the elevation **60**, whereby the receiving bore **62** is sealed towards the outside. On the side opposite the sealing ring **70** and on the free end face of the pressure equalization plug **64**, there is a membrane **74** as an example of a device **72** for closing the pressure equalization channel **66** in an air-permeable but water-impermeable manner, which membrane **74** is welded to the pressure equalization plug **64** in the embodiment example shown. The pressure equalization plug **64** is made of plastic in the present case.

[0031] In FIGS. 2 and 3, reference sign **76** indicates a cover cap covering the diaphragm, which is connected to the pressure equalization plug **64**. As illustrated in particular in FIG. 3, an equalizing channel **78** remains between the outer surfaces of the pressure equalizing plug **64** and the diaphragm **74**, which extends between an orifice **80** of the pressure equalizing channel **66** and an end of the cover cap **76** facing the elevation **60**. The cover cap **76** thus permits mechanical protection of the diaphragm **74**, but at the same time also pressure equalization between the inside of the housing **44** and the outside.

Claims

1. An electrical heating device for a motor vehicle, the electrical heating device comprising: a housing which encloses a heating chamber which is coupled in a thermally conductive manner to an electrical resistance element, the housing enclosing a control device; and a pressure equalization device for equalizing a pressure difference between the interior of the housing and the environment of the housing, which pressure equalization device is inserted in a sealing manner in a receiving bore which is designed to be adapted to receive the pressure equalization device and which is recessed in the housing.
 2. The electrical heating device according to claim 1, wherein the pressure compensation device is screwed into the receiving bore.
 3. The electrical heating device according to claim 2, wherein the pressure equalization device has a pressure equalization plug through which a pressure equalization channel passes, wherein the equalization channel is closed by an air-permeable but water-impermeable device.
 4. The electrical heating device according to claim 3, wherein the water-impermeable device comprises a diaphragm which is connected to the pressure compensation plug.
 5. The electrical heating device according to claim 4, wherein the diaphragm is connected to the pressure compensation plug with a material bond.
 6. The electrical heating device according to claim 4, wherein the pressure equalizing device has a cover cap which covers the diaphragm and is connected to the pressure equalizing plug.
 7. The electrical heating device according to claim 4, wherein the membrane is formed from PTFE and has a thickness of between 120 μm and 240 μm .
 8. The electrical heating device according to claim 7, wherein the membrane has a thickness of between 150 μm and 190 μm .
 9. The electrical heating device according to claim 6, wherein the diaphragm is made of metal.
 10. The electrical heating device according to claim 1, wherein the pressure equalizing device comprises a sealing ring, and wherein the pressure equalizing device and the sealing ring are integrally formed.
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